EP8199 - GLOBAL OPTIMIZATION OF MIXED-INTEGER AND NON-CONVEX PROGRAMS
COLLOQUIUM 3

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Outline

- RECAP:
- LOWER BOUNDING TECHNIQUES
- MISCELLANEOUS

Procedure for Global Optimization I

The general procedure is to determine a sequence of **Lower Bounds LBDs** and **Upper Bounds UBDs** that bracket the optimal value. Converge the best known UBD and LBD such that the gap **UBD - LBD** is within a tolerance. Solvers usually report this gap.

Upper Bounding Techniques:

1. Determine any **feasible** point. Value of objective provides an UBD.
2. **Restriction**: Shrink the feasible set of an optimization problem by constraining it further. The optimal value or value or objective function value at any feasible point yield an UBD for the original optimization problem.

Lower Bounding Techniques:

1. **Relaxation** of constraints
2. **Convexification**
3. Using **Duality theory**
4. **Interval Analysis**

Procedure for Global Optimization II

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The 2 most principles: Enlarging the feasible set of an optimization problem by relaxing constraints yields a lower bound. This is termed the Lower Bounding Principle:

Theorem (Lower Bounding Principle)

Suppose we have sets D and $E \subset \mathbb{R}^n$ such that $D \subset E$. Then

$$\min\{f(\mathbf{x}) : \mathbf{x} \in E\} \leq \min\{f(\mathbf{x}) : \mathbf{x} \in D\}$$

Theorem (Feasibility Principle - Prop. 2.11 Barton Book P.20)

Let $D, E \subset \mathbb{R}^n$ and suppose that $D \subset E$.

1. *If $\min_{\mathbf{x} \in E} f(\mathbf{x})$ is infeasible then $\min_{\mathbf{x} \in D} f(\mathbf{x})$ is infeasible.*
2. *Otherwise, if $\mathbf{x}^*$ is an optimal solution for $\min_{\mathbf{x} \in E} f(\mathbf{x})$ and $\mathbf{x}^* \in D$ then $\mathbf{x}^*$ is also an optimal solution for $\min_{\mathbf{x} \in D} f(\mathbf{x})$.*

1. Relaxation of Constraints

Determine the minimum of a relaxed problem (direct application of the feasibility principle). Has to be the minimum!

2. Convexification

This is the procedure of formulating an convex optimization problem that provides a LBD for a nonconvex problem

Definition (Convex Underestimator and Concave Overestimator of a function)

Consider a function $f : C \to \mathbb{R}$ defined on a convex set C. A convex function $f^{cv} : C \to \mathbb{R}$ and a concave function $f^{cc} : C \to \mathbb{R}$ are a convex underestimator and concave overestimator respectively of f on C if:

$$f^{cv}(\mathbf{x}) \leq f(\mathbf{x}), \quad \forall \mathbf{x} \in C$$
$$f^{cc}(\mathbf{x}) \geq f(\mathbf{x}), \quad \forall \mathbf{x} \in C$$

Together, these two functions are also referred to as convex and concave relaxations of a function. When these functions are the tightest possible relaxations, they are termed convex and concave envelopes.

- Library of convex relaxations for common univariate and bivariate functions. Active research area: Show Kannan Pg. 133.
- Building blocks to construct convex relaxations of more complicated functions.

Convexification - Auxiliary Variables and Constraints Method (AVM) I

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Constructing a convex programming relaxation for the following optimization problem:

$$\begin{aligned} \min_{x_1,x_2,x_3} \quad & f(x_1,x_2,x_3) = x_1 \cdot x_2 \cdot (5 + x_3^2) \\ \text{s.t.} \quad & x_1,x_2,x_3 \in [0,1] \end{aligned} \tag{1}$$

1. Break up the participating functions into a sequence of univariate and bivariate functions (also termed the factorable representation) by defining auxiliary variables. Determine the upper and lower bounds for the auxiliary variables based on the corresponding bounds for the original variables.

$$\begin{aligned} v_1 &= x_3^2 && \in [0,1] \\ v_2 &= 5 + v_1 && \in [5,6] \\ v_3 &= x_1 \cdot x_2 && \in [0,1] \\ v_4 &= v_2 \cdot v_3 && \in [0,6] \end{aligned}$$

Convexification - Auxiliary Variables and Constraints Method (AVM) II

2. Formulate an equivalent optimization problem in terms of these introduced variables and equality constraints:

$$
\begin{aligned}
\min_{x_1,x_2,x_3,v_1,v_2,v_3,v_4} \quad & v_4 \\
\text{s.t.} \quad & v_1 = x_3^2 \\
& v_2 = 5 + v_1 \\
& v_3 = x_1 \cdot x_2 \\
& v_4 = v_2 \cdot v_3 \\
& x_1, x_2, x_3 \in [0,1], v_1 \in [0,1], v_2 \in [5,6], v_3 \in [0,1], v_4 \in [0,6]
\end{aligned}
\tag{2}
$$

3. The above problem is still nonconvex as a result of the presence of nonlinear equality constraints. Thus, the last step is to replace each nonlinear equality constraint with a pair of inequality constraints in terms of their corresponding convex underestimators and concave overestimators:
   - For $v_1 = x_3^2$, it is known that $v_1^{cv} = x_3^2$ and $v_1^{cc} = x_3$ over $[0,1]$, Thus:
     $v_1 \geq x_3^2$ and $v_1 \leq x_3$
   - Function v_2 is affine thus convex and doesn't require reformulation

Convexification - Auxiliary Variables and Constraints Method (AVM) III

- For $v_3 = x_1 \cdot x_2$, the well-known McCormick envelopes can be used (where the superscripts L and U denote the lower and upper bound of the corresponding variable as presented in Problem 2):

$v_3^{cv} = \max\{x_1^L \cdot x_2 + x_2^L \cdot x_1 - x_1^L \cdot x_2^L,\ \ x_1^U \cdot x_2 + x_2^U \cdot x_1 - x_1^U \cdot x_2^U\}$

$v_3^{cc} = \min\{x_1^L \cdot x_2 + x_2^U \cdot x_1 - x_1^L \cdot x_2^U,\ \ x_1^U \cdot x_2 + x_2^L \cdot x_1 - x_1^U \cdot x_2^L\}$

This gives the following constraints after substitution of bounds:

$v_3 \geq 0$

$v_3 \geq x_2 + x_1 - 1$

$v_3 \leq x_1$

$v_3 \leq x_2$

- For $v_4 = v_2 \cdot v_3$, a similar procedure using McCormick envelopes gives:

$v_4 \geq 5 \cdot v_3$

$v_4 \geq 6 \cdot v_3 + v_2 - 6$

$v_4 \leq 5 \cdot v_3 + v_2 - 5$

$v_4 \leq 6 \cdot v_3$

Convexification - Auxiliary Variables and Constraints Method (AVM) IV

4. Substituting all the above inequalities yields Problem 3 which is a convex program:

$$
\begin{aligned}
\min_{x_1,x_2,x_3,v_1,v_2,v_3,v_4} \quad & v_4 \\
\text{s.t.} \quad & v_1 \geq x_3^2 \\
& v_1 \leq x_3 \\
& v_2 = 5 + v_1 \\
& v_3 \geq 0 \\
& v_3 \geq x_2 + x_1 - 1 \\
& v_3 \leq x_1 \\
& v_3 \leq x_2 \\
& v_4 \geq 5 \cdot v_3 \\
& v_4 \geq 6 \cdot v_3 + v_2 - 6 \\
& v_4 \leq 5 \cdot v_3 + v_2 - 5 \\
& v_4 \leq 6 \cdot v_3 \\
& x_1, x_2, x_3 \in [0,1], v_1 \in [0,1], v_2 \in [5,6], v_3 \in [0,1], v_4 \in [0,6]
\end{aligned}
\tag{3}
$$

Convexification - Auxiliary Variables and Constraints Method (AVM) V

NOTE:

- Using the AVM approach, the convex programming relaxation is typically of a much higher dimension.
- The optimal objective function value of the convex programming relaxation (e.g., Problem 3) is a lower bound on the optimal objective function value of the nonconvex program (e.g., Problem 1).

3. Using Duality Theory I

Consider the NLP Problem NLP-PR which is also termed the primal.

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{g}(\mathbf{x}) \leq \mathbf{0} \\ & \mathbf{x} \in X \subset \mathbb{R}^{n_x} \end{aligned} \tag{NLP-PR}$$

Definition (Lagrange function for Problem NLP-PR)

The Lagrange function (L) adds the objective function with a weighted sum of the constraint functions i.e., $L : \mathbb{R}^{n_x} \times \mathbb{R}^m \to \mathbb{R}$ is given by:

$$L(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \boldsymbol{\lambda}^{\mathrm{T}} \mathbf{g}(\mathbf{x}) \tag{4}$$

Each element of the vector $\boldsymbol{\lambda}$ (termed a Lagrange multiplier) corresponds to a constraint.

3. Using Duality Theory II

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Definition (The Dual function for Problem NLP-PR)

The dual function q is defined implicitly as the optimal objective function value of an optimization problem parameterized by a Lagrange multiplier vector, i.e., $q : \mathbb{R}^m \to \mathbb{R}$ is given by:

$$q(\boldsymbol{\lambda}) = \inf_{\mathbf{x} \in X} L(\mathbf{x}, \boldsymbol{\lambda}) \tag{5}$$

We note that q is only defined on $\boldsymbol{\lambda}$ for which the optimization problem above is bounded.

Definition (The Dual of Problem NLP-PR)

The Dual is an optimization problem defined as:

$$\begin{aligned} \sup_{\boldsymbol{\lambda} \in \mathbb{R}^m} \quad & q(\boldsymbol{\lambda}) \\ \text{s.t.} \quad & \boldsymbol{\lambda} \geq \mathbf{0}, \end{aligned} \tag{NLP-DP-DEF}$$

3. Using Duality Theory III

The above definitions are combined to write out an alternative concise Dual Problem (Problem NLP-DP)

Definition (Dual - concise definition)

$$\sup_{\boldsymbol{\lambda} \geq \mathbf{0}} \quad \inf_{\mathbf{x} \in X} f(\mathbf{x}) + \boldsymbol{\lambda}^{\mathrm{T}} \mathbf{g}(\mathbf{x}) \tag{NLP-DP}$$

Definition (A Duality Multiplier of Problem NLP-PR)

A vector $\hat{\boldsymbol{\lambda}}$ is said to be a Duality Multiplier of Problem NLP-PR if:

$$\hat{\boldsymbol{\lambda}} \geq \mathbf{0} \quad \text{and} \quad f(\hat{\mathbf{x}}) = \inf_{\mathbf{x} \in X} f(\mathbf{x}) + (\hat{\boldsymbol{\lambda}})^{\mathrm{T}} \mathbf{g}(\mathbf{x}) \tag{6}$$

where $\hat{\mathbf{x}}$ in this case denotes the globally optimal solution. We note that the duality multiplier is not the same as a Lagrange multiplier and may not exist for certain problem classes.

3. Using Duality Theory IV

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Theorem (Weak duality)

For all problem classes, the supremum on the Dual Problem NLP-DP is a lower bound on the optimal objective function value of the primal NLP (Problem NLP-PR).

Theorem (Strong duality and existence of duality multiplier for convex programs)

Suppose Problem NLP-PR is a convex program and has a finite solution value. Also suppose a strictly feasible point exists (also called a Slater point - a feasible point where none of the constraints are active). Then strong duality holds and Problems NLP-DP and NLP-PR have the same optimum objective function value. In addition, atleast one duality multiplier exists.

Exercise : Please recall the approach Paul presented on the viewing duality through the image space

Why notions of open, closed set, boundedness were introduced

1. Differentiability is usually defined on an open set
2. **Existence of minima: Weierstrass theorem**

Theorem (Weierstrass Theorem)

If $D \subset \mathbb{R}^n$ is a nonempty compact set and $f : D \to \mathbb{R}$ is continuous on D, then both a minimum and a maximum of f on D exist.